David Simons

BSc, MBBS, DTM&H, MSc, PhD

Birgitta Sintring Fellow, Department of Ecology and Genetics, Uppsala University

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APPOINTMENTS

Birgitta Sintring Independent Research Fellow Sept 2026 – present
Department of Ecology and Genetics, Uppsala University, Sweden

Collaborator, SCAPES project 2026 – present
The Pennsylvania State University, USA

External Expert 2025 – present
European Centre for Disease Prevention and Control (ECDC), Stockholm, Sweden

Postdoctoral Scholar 2023 – 2026
Department of Anthropology and Center for Infectious Disease Dynamics, The Pennsylvania State University, USA

Fellow-in-Residence 2023 – present
The Verena Institute, USA

Senior Analyst, Outbreak Surveillance Team 2021 – 2022
UK Health Security Agency, London, UK

Visiting Researcher 2020
EcoHealth Alliance, New York, USA

Resident Doctor, Hospital for Tropical Diseases 2019 – 2022
University College London Hospitals, London, UK

Locum Senior House Officer 2018 – 2019
Various NHS hospitals, London, UK

Foundation Doctor 2016 – 2018
North-East London Foundation Training School, London, UK

Research Assistant, Department of Microbiology, Tumour and Cell Biology 2012
Karolinska Institutet, Stockholm, Sweden

Clinical Research Scientist (industrial placement) 2008 – 2009
GlaxoSmithKline, Harlow, UK

EDUCATION

PhD, Epidemiology of Emerging Zoonotic Infectious Diseases 2024
University of London (RVC, UCL and LSHTM), London, UK

MSc, Tropical Medicine and International Health 2019
London School of Hygiene and Tropical Medicine, London, UK

Professional Diploma in Tropical Medicine and Hygiene (DTM&H) 2018
London School of Hygiene and Tropical Medicine, London, UK

Membership of the Royal College of Physicians (Part 1) 2017

Royal College of Physicians, London, UK

MBBS, Bachelor of Medicine and Bachelor of Surgery 2016
King's College London, GKT School of Medical Education, London, UK

MSc and BSc, Natural Sciences 2011
University of Leeds, Leeds, UK

RESEARCH FUNDING

Funded

Birgitta Sintring Fellowship, Principal Investigator 2026 – present
Functional connectivity and microbial transmission pathways in Fennoscandian rodents. Uppsala University.

Vetenskapsrådet, Sweden–South Korea Research Collaboration, Principal Investigator 2026 – 2027
An AI-directed One Health risk prediction framework for active learning in rodent-borne zoonoses. Uppsala University, with Chungnam National University.

Formas, CASCADE, Co-applicant and work package lead (modelling), 50% of full time 2026 – 2030
Climate-amplified spillover of vector-borne and zoonotic diseases in Sweden. Project leader Hussein Khalil, SLU. Nearly SEK 9,000,000 total.

Verena Institute, Co-Principal Investigator 2023 – 2024
Project ArHa: arenaviruses and hantaviruses. $5,000.

EcoHealth Alliance, Principal Investigator 2020
Forecasting global veterinary disease: modelling African swine fever. $5,000.

BBSRC LIDo Doctoral Training Partnership, PhD studentship 2019 – 2024
Investigating rodent ecology to understand Lassa fever spillover. £85,000.

EDITORIAL AND PROFESSIONAL SERVICE

Academic Editor, *PLOS Neglected Tropical Diseases* Feb 2026 – present

E-learning curriculum, ECDC 2025
Co-developed an e-learning course on the environmental determinants of communicable diseases.

Peer review
26 reviews for 14 international journals; 2 book chapter reviews.

Grant review
Grant Agency of the Czech Republic.

Scientific committee, Annual Rodent-Borne Zoonotic Disease Workshop

TEACHING, MENTORING AND SUPERVISION

Thesis supervision 2019 – present
Joint or sole supervisor of 8 students on medical elective, undergraduate and postgraduate research projects at UCL, LSHTM and the Royal Veterinary College.

Postgraduate and professional lecturing 2021 – present
Invited lecturer on postgraduate courses at LSHTM (DTM&H) and UCL (Infectious Diseases MSc).

Undergraduate lecturing 2023 – present
"Spillover: The Anthropology of Emerging Infectious Diseases", The Pennsylvania State University.

Field and laboratory capacity building 2019 – present

Recruited, trained and supervised field and laboratory teams in Sierra Leone and Nigeria in rodent sampling, PCR and antibody assays (ELISA, LIPS) under BSL-3 conditions.

INVITED PRESENTATIONS

36 invited talks and posters at international conferences and institutions, including the ECOWAS Lassa Fever International Conference, ASTMH, EEID, LSHTM and the University of Idaho. Full list on request.

SKILLS

Data analysis and programming. R (tidyverse, Quarto, Shiny, Stan, sf), QGIS, ArcGIS, Git and GitHub, SQL.

Laboratory and field. Rodent trapping and handling, serological assays (ELISA), molecular diagnostics (PCR).

Clinical. Infectious diseases, tropical medicine, emergency medicine. Registered with the General Medical Council.

Languages. English (native), Swedish (learning).

PUBLICATIONS

68 papers and preprints, newest first. Full list with links at dsimons.org/publications.

68. Imirzian N, Trebski A, Johnson E, Hewitson M, **Simons D**, Harden C, Friant S, Redding D (2026). 'SiteTool': a 'Shiny' application for field site selection and evaluation. *Ecography*.

67. Simons D, Watson-Jones D, Manno D, Friant S (2026). A One Health trial design to accelerate Lassa fever vaccines. *Nature Medicine*.

66. Catchen MD, Banville F, Boutin AC, Brookson CB, Carlson CJ, Dansereau G, …, **Simons D**, et al. (2026). A protocol for biodiversity-informed wildlife disease surveillance. *PLOS Ecosystems*.

65. Simons D, Rivero R, Rickard G, Guiote A, Gordon H, Milne G, Redding D, Seifert S (2026). ArHa: A globally standardised, relational database of arenavirus and hantavirus sampling in wild small mammal populations. Preprint.

64. Simons D, Goyal R, Bangura U, Gibb R, Rushton B, Sondufu D, et al. (2026). Contact Networks of Small Mammals Highlight Potential Transmission Foci of Mammarenavirus lassaense. *The American Journal of Tropical Medicine and Hygiene*.

63. Simons D, Harden C, Imirzian N, Thompson KET, Ifebueme NM, Eziechina S, et al. (2026). Local heterogeneity in Lassa fever serology in rural Nigeria: Implications for vaccine trial site selection. *PLOS Neglected Tropical Diseases*.

62. Harden C, Eziechina S, Ifebueme NM, **Simons D**, Moses L, Redding DW, Friant S (2026). Local Knowledge of Land Use Effects on Human-Rodent Interfaces in Lassa-Endemic Nigeria. *EcoHealth*.

61. Simons D (2026). Map Liberator: An open-source tool for recovering spatial epidemiological data from static situation reports. *medRxiv*. Preprint.

60. Rivero R, **Simons D**, Damodaran L, Karegi I, Gurev S, Becker DJ, et al. (2026). Molecular and ecological determinants of effective reassortment in orthohantaviruses. *bioRxiv*. Preprint.

59. Redding DW, Albaladejo-Robles G, Nino Barreat JG, **Simons D**, Abbate JL (2026). Open ecological, genomic and digital behavioural data resolve the 2026 MV Hondius Andes virus outbreak to the southern-Andean reservoir zone. *medRxiv*. Preprint.

58. Leppin C, Brown J, Garnett C, Kale D, Okpako T, **Simons D**, Perski O (2026). Optimising supervised machine learning algorithms predicting cigarette cravings and lapses for a smoking cessation just-in-time adaptive intervention (JITAI). *PLOS One*.

57. Simons D (2026). Reported case fatality as a target and trial metric in endemic zoonoses: a framework and ecological analysis of Lassa fever surveillance in Nigeria, 2017–2026. *medRxiv*. Preprint.

56. Friant S, **Simons D**, Harden C, Imirzian N, Bjornstad O, Gibb R, et al. (2026). Scaling Lassa Virus Dynamics within Anthropogenic Ecosystems (SCAPES) Study Protocol: a mixed-methods observational cohort study of humans, rodents, and landscapes in Nigeria. *Wellcome Open Research*.

55. Simons D (2026). The socio-economic shield limits Lassa virus spillover in urban West Africa. *Epidemiology and Infection*.

54. Perski O, Meinecke F, Roberto MS, Stenfors CUD, Carlbring P, **Simons D** (2026). Use of Global Positioning Systems to detect, predict and intervene on addictive behaviours: A systematic review. Preprint.

53. Simons D, Rivero R, Rickard G, Martinez-Checa A, Gordon H, Redding DW, Seifert SN (2026). Viral reservoir status in small mammals emerges as a predictable life-history trait after correcting for surveillance bias. *bioRxiv*. Preprint.

52. Schwantes CJ, Sánchez CA, Stevens T, Zimmerman R, Albery G, Becker DJ, …, **Simons D**, et al. (2025). A minimum data standard for wildlife disease research and surveillance. *Scientific Data*.

51. Simons D, Gibb R, Bangura U, Sondufu D, Lamin J, Koninga J, et al. (2025). Land use gradients drive spatial variation in Lassa fever host communities in the Eastern Province of Sierra Leone. *Journal of Animal Ecology*.

50. Simons D, Rivero R, Guiote AMC, Gordon HLM, Milne GC, Rickard G, Redding DW, Seifert SN (2025). Protocol to produce a systematic Arenavirus and Hantavirus host-pathogen database: Project ArHa. *Wellcome Open Research*.

49. Friant S, Mistrick J, Luis AD, Harden C, **Simons D**, Fichet-Calvet E, et al. (2025). Reducing the threats of rodent-borne zoonoses requires an understanding and leveraging of three key pillars: disease ecology, synanthropy, and rodentation. *The Lancet Planetary Health*.

48. Streicker DG, Griffiths ME, Antia R, Bergner L, Bowman P, de Moraes MVdS, …, **Simons D**, et al. (2024). Developing transmissible vaccines for animal infections. *Science*.

47. Perski O, Kale D, Leppin C, Okpako T, **Simons D**, Goldstein SP, Hekler E, Brown J (2024). Supervised machine learning to predict smoking lapses from Ecological Momentary Assessments and sensor data: Implications for just-in-time adaptive intervention development. *PLOS Digital Health*.

46. Simons D, Attfield LA, Jones KE, Watson-Jones D, Kock R (2023). A dataset of small-mammal detections in West Africa and their associated micro-organisms. *Gigabyte*.

45. Perski O, Li K, Pontikos N, **Simons D**, Goldstein SP, Naughton F, Brown J (2023). Classification of Lapses in Smokers Attempting to Stop: A Supervised Machine Learning Approach Using Data From a Popular Smoking Cessation Smartphone App. *Nicotine and Tobacco Research*.

44. Allen H, Tessier E, Turner C, Anderson C, Blomquist P, **Simons D**, et al. (2023). Comparative transmission of SARS-CoV-2 Omicron (B.1.1.529) and Delta (B.1.617.2) variants and the impact of vaccination: national cohort study, England. *Epidemiology and Infection*.

43. Arruda LB, Free HB, **Simons D**, Ansumana R, Elton L, Haider N, et al. (2023). Current sampling and sequencing biases of Lassa mammarenavirus limit inference from phylogeography and molecular epidemiology in Lassa fever endemic regions. *PLOS Global Public Health*.

42. Simons D, Attfield LA, Jones KE, Watson-Jones D, Kock R (2023). Rodent trapping studies as an overlooked information source for understanding endemic and novel zoonotic spillover. *PLOS Neglected Tropical Diseases*.

41. Simons D, Perski O, Shahab L, Brown J, Bailey R (2022). Association of smoking status with hospitalisation for COVID-19 compared with other respiratory viruses a year previous: a case-control study at a single UK National Health Service trust. *F1000Research*.

40. Vivancos R, Anderson C, Blomquist P, Balasegaram S, Bell A, Bishop L, …, **Simons D**, et al. (2022). Community transmission of monkeypox in the United Kingdom, April to May 2022. *Eurosurveillance*.

39. Andrews N, Stowe J, Kirsebom F, Toffa S, Rickeard T, Gallagher E, …, **Simons D**, et al. (2022). Covid-19 Vaccine

Effectiveness against the Omicron (B.1.1.529) Variant. *New England Journal of Medicine*.

38. Kirsebom FCM, Andrews N, Stowe J, Toffa S, Sachdeva R, Gallagher E, et al. (2022). COVID-19 vaccine effectiveness against the omicron (BA.2) variant in England. *The Lancet Infectious Diseases*.

37. Blomquist PB, Bridgen J, Bray N, O'Connell AM, West D, Groves N, …, **Simons D**, et al. (2022). Enhancing epidemiological surveillance of the emergence of the SARS-CoV-2 Omicron variant using spike gene target failure data, England, 15 November to 31 December 2021. *Eurosurveillance*.

36. Haider N, Guitian J, **Simons D**, Asogun D, Ansumana R, Honeyborne I, et al. (2022). Increased outbreaks of monkeypox highlight gaps in actual disease burden in Sub-Saharan Africa and in animal reservoirs. *International Journal of Infectious Diseases*.

35. Simons D (2022). Lassa fever cases suffer from severe underreporting based on reported fatalities. *International Health*.

34. Barnard RC, Davies NG, Centre for Mathematical Modelling of Infectious Diseases COVID-19 working group, Munday JD, Lowe R, Knight GM, …, **Simons D**, et al. (2022). Modelling the medium-term dynamics of SARS-CoV-2 transmission in England in the Omicron era. *Nature Communications*.

33. Nightingale ES, Abbott S, Russell TW, CMMID Covid-19 Working Group, Rees EM, Eggo RM, …, **Simons D**, et al. (2022). The local burden of disease during the first wave of the COVID-19 epidemic in England: estimation using different data sources from changing surveillance practices. *BMC Public Health*.

32. Leclerc QJ, Nightingale ES, Abbott S, CMMID COVID-19 Working Group, Rees EM, Eggo RM, …, **Simons D**, et al. (2021). Analysis of temporal trends in potential COVID-19 cases reported through NHS Pathways England. *Scientific Reports*.

31. Osman AY, Elmi SA, **Simons D**, Elton L, Haider N, Khan MA, et al. (2021). Antimicrobial Resistance Patterns and Risk Factors Associated with Salmonella spp. Isolates from Poultry Farms in the East Coast of Peninsular Malaysia: A Cross-Sectional Study. *Pathogens*.

30. Davis EL, Lucas TCD, Borlase A, Pollington TM, Abbott S, Ayabina D, …, **Simons D**, et al. (2021). Contact tracing is an imperfect tool for controlling COVID-19 transmission and relies on population adherence. *Nature Communications*.

29. The SAFER Investigators and Field Study Team, Hellewell J, The Crick COVID-19 Consortium, CMMID COVID-19 working group, Russell TW, Beale R, et al. (2021). Estimating the effectiveness of routine asymptomatic PCR testing at different frequencies for the detection of SARS-CoV-2 infections. *BMC Medicine*.

28. Blunsum AE, Perkins JS, Arshad A, Bajpai S, Barclay-Elliott K, Brito-Mutunayagam S, …, **Simons D**, et al. (2021). Evaluation of the Implementation of the 4C Mortality Score in United Kingdom hospitals during the second pandemic wave. *medRxiv*. Preprint.

27. Elmi SA, **Simons D**, Elton L, Haider N, Abdel Hamid MM, Shuaib YA, et al. (2021). Identification of Risk Factors Associated with Resistant Escherichia coli Isolates from Poultry Farms in the East Coast of Peninsular Malaysia: A Cross Sectional Study. *Antibiotics*.

26. Munday JD, Sherratt K, Meakin S, Endo A, Pearson CAB, Hellewell J, …, **Simons D**, et al. (2021). Implications of the school-household network structure on SARS-CoV-2 transmission under school reopening strategies in England. *Nature Communications*.

25. Leclerc QJ, Fuller NM, Keogh RH, Diaz-Ordaz K, Sekula R, Semple MG, …, **Simons D**, et al. (2021). Importance of patient bed pathways and length of stay differences in predicting COVID-19 hospital bed occupancy in England. *BMC Health Services Research*.

24. Prem K, Zandvoort Kv, Klepac P, Eggo RM, Davies NG, Centre for the Mathematical Modelling of Infectious Diseases COVID-19 Working Group, Cook AR, Jit M (2021). Projecting contact matrices in 177 geographical regions: An update and comparison with empirical data for the COVID-19 era. *PLOS Computational Biology*.

23. Quilty BJ, Clifford S, Hellewell J, Russell TW, Kucharski AJ, Flasche S, …, **Simons D**, et al. (2021). Quarantine and testing strategies in contact tracing for SARS-CoV-2: a modelling study. *The Lancet Public Health*.

22. Perski O, **Simons D**, Shahab L, Brown J (2021). Smoking, Nicotine, and COVID-19: Triangulation of Methods and Preregistration Are Required for Robust Causal Inference. *Nicotine & Tobacco Research*.

21. Clifford S, Quilty BJ, Russell TW, Liu Y, Chan YWD, Pearson CAB, et al. (2021). Strategies to reduce the risk of SARS-CoV-2 importation from international travellers: modelling estimations for the United Kingdom, July 2020. *Eurosurveillance*.

20. CMMID COVID-19 working group, Jarvis CI, Gimma A, van Zandvoort K, Wong KLM, Edmunds WJ (2021). The impact of local and national restrictions in response to COVID-19 on social contacts in England: a longitudinal natural experiment. *BMC Medicine*.

19. CMMID COVID-19 Working Group, Liu Y, Morgenstern C, Kelly J, Lowe R, Jit M (2021). The impact of non-pharmaceutical interventions on SARS-CoV-2 transmission across 130 countries and territories. *BMC Medicine*.

18. Nightingale ES, Brady OJ, CMMID Covid-19 working group, Yakob L (2021). The importance of saturating density dependence for population-level predictions of SARS-CoV-2 resurgence compared with density-independent or linearly density-dependent models, England, 23 March to 31 July 2020. *Eurosurveillance*.

17. LSHTM CMMID COVID-19 Working Group, Mburu CN, Ojal J, Chebet R, Akech D, Karia B, et al. (2021). The importance of supplementary immunisation activities to prevent measles outbreaks during the COVID-19 pandemic in Kenya. *BMC Medicine*.

16. Arruda LB, Haider N, Olayemi A, **Simons D**, Ehichioya D, Yinka-Ogunleye A, et al. (2021). The niche of One Health approaches in Lassa fever surveillance and control. *Annals of Clinical Microbiology and Antimicrobials*.

15. Sandmann FG, Davies NG, Vassall A, Edmunds WJ, Jit M, Sun FY, …, **Simons D**, et al. (2021). The potential health and economic value of SARS-CoV-2 vaccination alongside physical distancing in the UK: a transmission model-based future scenario analysis and economic evaluation. *The Lancet Infectious Diseases*.

14. Gibbs H, Liu Y, Pearson CAB, Jarvis CI, Grundy C, Quilty BJ, …, **Simons D**, et al. (2020). Changing travel patterns in China during the early stages of the COVID-19 pandemic. *Nature Communications*.

13. Chevance G, Hekler EB, Efoui-Hess M, Godino J, Golaszewski N, Gualtieri L, …, **Simons D**, et al. (2020). Digital health at the age of the Anthropocene. *The Lancet Digital Health*.

12. Kucharski AJ, Klepac P, Conlan AJK, Kissler SM, Tang ML, Fry H, …, **Simons D**, et al. (2020). Effectiveness of isolation, testing, contact tracing, and physical distancing on reducing transmission of SARS-CoV-2 in different settings: a mathematical modelling study. *The Lancet Infectious Diseases*.

11. Perski O, **Simons D**, West R, Michie S (2020). Face masks to prevent community transmission of viral respiratory infections: A rapid evidence review using Bayesian analysis. Preprint.

10. Kapata N, Ihekweazu C, Ntoumi F, Raji T, Chanda-Kapata P, Mwaba P, …, **Simons D**, et al. (2020). Is Africa prepared for tackling the COVID-19 (SARS-CoV-2) epidemic. Lessons from past outbreaks, ongoing pan-African public health efforts, and implications for the future. *International Journal of Infectious Diseases*.

9. Dexter D, **Simons D**, Kiyaga C, Kapata N, Ntoumi F, Kock R, Zumla A (2020). Mitigating the effect of the COVID-19 pandemic on sickle cell disease services in African countries. *The Lancet Haematology*.

8. Haider N, Yavlinsky A, **Simons D**, Osman AY, Ntoumi F, Zumla A, Kock R (2020). Passengers' destinations from China: low risk of Novel Coronavirus (2019-nCoV) transmission into Africa and South America. *Epidemiology and Infection*.

7. Russell TW, Golding N, Hellewell J, Abbott S, Wright L, Pearson CAB, …, **Simons D**, et al. (2020). Reconstructing the early global dynamics of under-ascertained COVID-19 cases and infections. *BMC Medicine*.

6. Simons D, Brown J, Shahab L, Perski O (2020). Smoking and COVID-19: Rapid evidence review for the Royal College of Physicians, London (UK). *Qeios*. Preprint.

5. Simons D, Shahab L, Brown J, Perski O (2020). The association of smoking status with SARS-CoV-2 infection, hospitalization and mortality from COVID-19: a living rapid evidence review with Bayesian meta-analyses (version 7).

Addiction.

4. Emery JC, Russell TW, Liu Y, Hellewell J, Pearson CA, CMMID COVID-19 Working Group, …, **Simons D**, et al. (2020). The contribution of asymptomatic SARS-CoV-2 infections to transmission on the Diamond Princess cruise ship. *eLife*.

3. Quaife M, van Zandvoort K, Gimma A, Shah K, McCreesh N, Prem K, …, **Simons D**, et al. (2020). The impact of COVID-19 control measures on social contacts and transmission in Kenyan informal settlements. *BMC Medicine*.

2. Firth JA, Hellewell J, Klepac P, Kissler S, CMMID COVID-19 Working Group, Jit M, …, **Simons D**, et al. (2020). Using a real-world network to model localized COVID-19 control strategies. *Nature Medicine*.

1. O'Neill BV, Bullmore ET, Miller S, McHugh S, **Simons D**, Dodds CM, et al. (2012). The relationship between fat mass, eating behaviour and obesity-related psychological traits in overweight and obese individuals. *Appetite*.